

Confidence Intervals

LLE – Mathematics and Statistics Skills

1. Let's say that a researcher is interested in the mean length of time people spend using a particular product each year. They collect a sample from the population of people who have the product and calculate the mean length to be 6.4 days.
 - (a) With the information given, what would be the researcher's estimate of the population mean number of days people are using the product?
 - (b) Using other statistics collected from the sample, the researcher also calculates a 95% confidence interval for the population mean. It is calculated as (4.6, 8.2).

What is the interpretation of this confidence interval?

- ☐ We are 5% confident that the true mean is between 4.6 and 8.2 days
 - ☐ We are 95% confident that the true mean is between 4.6 and 8.2 days
 - ☐ We are 95% confident that the true mean is outside of 4.6 and 8.2 days
 - ☐ We have no idea how many days people use the product for for on average
2. The margin of error in a confidence interval is determined by the level of confidence, the sample size, and the standard deviation. When the population standard deviation is known you can use the formula, for a 95% confidence interval:

$$\bar{x} \pm 1.96 \times \frac{\sigma}{\sqrt{n}}$$

Where $\bar{x}$ is the sample mean, σ is the population standard deviation, n is the sample size.

- (a) Calculate the 95% confidence interval if sample mean = 40, population standard deviation = 10, sample size = 100
 - (b) Calculate the 95% confidence interval if sample mean = 100, population standard deviation = 15, sample size = 25
 - (c) Calculate the 95% confidence interval if sample mean = 100, population standard deviation = 15, sample size = 225
 - (d) Calculate the 95% confidence interval if sample mean = 100, population standard deviation = 75, sample size = 225
3. Other size confidence intervals can be calculated by changing the 1.96 part of the formula. The correct value can be obtained from normal tables.
- (a) What is the critical value for a 99% confidence interval of the mean? Give answer to 3 d.p. and positive.
 - (b) Given a sample mean of 100, population standard deviation of 30, and sample size of 225, what is the 99% confidence interval of the mean?
4. If you do not have the population standard deviation when calculating a confidence interval for the mean, you can make use of the standard deviation of the sample. When doing this you will need to make use of the t-distribution, rather than the normal distribution.
- From a sample of size 20, a sample mean of 156 and a sample standard deviation of 12.2 is found.
- (a) What would be the critical t value for a 95% confidence interval, correct to 3 d.p.?
 - (b) What would be the 95% confidence interval for the mean?

5. From a sample of size 12, a sample mean of 100 and a sample standard deviation of 15 is found.

What is the 90% confidence interval for the mean, correct to 2 d.p.?

6. (a) Complete the table of confidence intervals for the sample statistics, correct to 1 decimal place.

Size	Mean	Std. Dev.	95% Conf. Int.	99% Conf. Int.
15	50	4		
15	50	15		
10	50	15		

- (b) What do you notice to the width (difference between upper and lower bounds) of the confidence interval as the confidence percent increases?
- ☐ As the confidence percent increases, the width of the interval decreases
 - ☐ As the confidence percent increases, the width of the interval increases
- (c) What do you notice to the width (difference between upper and lower bounds) of the confidence interval as the sample size increases?
- ☐ As the sample size increases, the width of the interval decreases
 - ☐ As the sample size increases, the width of the interval increases
7. A class from the Environmental Science School collects 20 samples of water from a lake. The mean pH value of the 20 samples is 8.2
The local council collects 100 samples from the lake and gets a mean pH of 8.
It is known that the standard deviation of the pH values for the lake is 0.6.
- (a) Calculate 95% confidence intervals for the mean pH from the samples from the environmental science class

- (b) Will the local council have a wider 95% confidence interval than the science class?
8. A random sample of 15 callers to the “Pecs Bar & Gym” support line showed that they were on hold for the following number of minutes:
- 19, 28, 34, 10, 27, 31, 25, 37, 54, 27, 54, 8, 17, 24, 25
- (a) What is the mean waiting time for the sample?
- (b) What is the standard deviation of waiting time for the sample, correct to 2 d.p.?
- (c) Given that the confidence interval is written as Mean $\pm$ Error, what is the value of the Error, to 2 d.p.?
9. A random sample of measurements is taken from a normal distribution with an unknown mean and a known population standard deviation of 15.
- (a) What is the smallest sample size so that the width of a 95% confidence interval is smaller than 6 units?
- (b) What is the smallest sample size so that the width of a 95% confidence interval is smaller than 3 units?